Hoang (Bolton) Tran, Ph.D.

Ann Arbor, MI | hoangtra@umich.com | +1 (267) 206-6705 | https://bolton2710.github.io/

Education

Pennsylvania State University, Ph.D. in Chemical Engineering — Sept 2018 – Aug 2023
Thesis: Electrocatalysis at the Electrode/Electrolyte Interface: a Multiscale Molecular Model
Advisors: Prof. Michael J. Janik & Prof. Scott T. Milner

Drexel University, B.S. in Chemical Engineering — Sept 2012 – June 2017

Research

Postdoctoral research fellow — Sept 2023 – present
University of Michigan, Department of Chemical Engineering
Advisor: Prof. Bryan R. Goldsmith

Graduate research assistant — Sept 2018 – Aug 2023
Pennsylvania State University, Department of Chemical Engineering
Advisors: Prof. Michael J. Janik & Prof. Scott T. Milner

Undergraduate research assistant — Sept 2014 – Apr 2015
Drexel University, Department of Chemical Engineering
Advisors: Prof. Jason Baxter

Industry

Avid Radiopharmaceuticals, Philadelphia, PA
Quality assurance associate — Nov 2016 – Aug 2018
Manufacturing intern — Mar 2015 – Aug 2016

Zeolyst International, Conshohocken, PA
Analytical intern — May 2014 – Sept 2014

Teaching

Atomistic scale simulations, ME/CHE 505, Pennsylvania State University — Spring 2022
Guest lecturer (six one-hour lectures)

Simulation techniques and applications, CHE 597, Pennsylvania State University — Fall 2022
Guest lecturer (a one-hour lecture)

Material balance, CHE 210, Pennsylvania State University — Fall 2022
Graduate teaching assistant

Reaction engineering, CHE 430, Pennsylvania State University — Spring 2021
Graduate teaching assistant

Research Mentoring

Pennsylvania State University

- Elizabeth Long, Women In Science and Engineering Research (WISER) undergraduate researcher
- Yusheng Cai, undergraduate researcher, now a Ph.D. candidate at the University of Pennsylvania
- Joe Hughes, undergraduate researcher, now an engineer at Naval Reactors

University of Michigan

- Ankit Mathanker, 4th year Ph.D. candidate
- Dean Sweeney, 2nd year Ph.D. candidate
- Roshini Dantuluri, 2nd year Ph.D. candidate
- Yifei Liu, undergraduate researcher
- Mad Lindsey, undergraduate researcher
- Jean-Pattrick Selo, community college researcher

Publications

Published peer-reviewed papers

1. **Tran**, **B.**, Janik, M. J., Milner, S. T., "Hydration-Shell Solvation and Screening Govern Alkali Cation Concentrations at Electrochemical Interfaces", J. Phys. Chem. C **128**, 20559–20568 (2024)
2. Mathanker, A., Halarnkar, S., **Tran**, **B.**, Singh, N., Goldsmith, B. R., "Synergistic effects in organic mixtures for enhanced catalytic hydrogenation and hydrodeoxygenation", Chem Catalysis, 101135 (2024)
3. **Tran**, **B.**, Goldsmith, B. R., "Theoretical Investigation of the Potential-Dependent CO Adsorption on Copper Electrodes", J. Phys. Chem. Lett. **15**, 6538–6543 (2024)
4. Wong, A. J., **Tran**, **B.**, Agrawal, N., Goldsmith, B. R., Janik, M. J., "Sensitivity Analysis of Electrochemical Double Layer Approximations on Electrokinetic Predictions: Case Study for CO Reduction on Copper", J. Phys. Chem. C **128**, 10837–10847 (2024)
5. **Tran**, **B.**, Zhou, Y., Janik, M. J., Milner, S. T., "Negative Dielectric Constant of Water at a Metal Interface", Phys. Rev. Lett. **131**, 248001 (2023)
6. Ostervold, L., Daneshpour, R., Facchinei, M., **Tran**, **B.**, Wetherington, M., Alexopoulos, K., Greenlee, L., Janik, M. J., "Identifying the Local Atomic Environment of the "Cu^{3+}" State in Alkaline Electrochemical Systems", ACS Appl. Mater. Interfaces **15**, 27878–27892 (2023)
7. **Tran**, **B.**, Milner, S. T., Janik, M. J., "Kinetics of Acid-Catalyzed Dehydration of Alcohols in Mixed Solvent Modeled by Multiscale DFT/MD", ACS Catal. **12**, 13193–13206 (2022)
8. **Tran**, **B.**, Cai, Y., Janik, M. J., Milner, S. T., "Hydrogen Bond Thermodynamics in Aqueous Acid Solutions: A Combined DFT and Classical Force-Field Approach", J. Phys. Chem. A **126**, 7382–7398 (2022)
9. Edley, M. E., Opasanont, B., Conley, J. T., **Tran**, **H.**, Smolin, S. Y., Li, S., Dillon, A. D., Fafarman, A. T., Baxter, J. B., "Solution Processed $CuSbS_2$ Films for Solar Cell Applications", Thin Solid Films **646**, 180–189 (2018)

Manuscripts in revision, submission, or preparation

1. Long, E., **Tran, B.**, Milner, S. T., "Tuning Partial Charges of Alkyl Alcohols to Improve Simulated Fluid Properties", J. Chem. Phys., manuscript in revision.
2. Wong, A. J., **Tran, B.**, Milner, S. T., Janik, M. J., "Electrode-Electrolyte Interfacial Effects on Specific Alkali Metal Cation Adsorption to Late Transition Metal Surfaces", manuscript in preparation.
3. Manthanker, A., Sharma, G., **Tran, B.**, Signh, N., Goldsmith, B. R., "Effect of Ions on the Aqueous-Phase Adsorption of Benzene, Phenol, and Catechol on Ag(111)", manuscript in preparation.
4. **Tran, B.**, Goldsmith, B. R., "Predicting Competitive Anion Electrosorption on Late Transition Metal Surfaces", manuscript in preparation.
5. **Tran, B.**, Sweeney, D., Selo, J., Liu, Y., Lindsey, M., Goldsmith, B. R., "Effects of hydration on the predictive power of O-H stretching frequency on acid pKa", manuscript in preparation.

Conference Presentation

1. **AIChE 2024** | San Diego | oral & poster presentations.
2. **ACS Fall 2024** | Denver | oral & poster presentations.
3. **GRC/GRS 2024** | New London | oral (invited) & poster presentations.
4. **Michigan Catalysis Society 2024** | Ann Arbor | oral presentation.
5. **ACS Spring 2023** | Indianapolis | oral presentation.
6. **AIChE 2022** | Phoenix | oral & poster presentations.
7. **Pittsburgh-Cleveland Catalysis Society 2022** | State College | poster presentation.
8. **North American Catalysis Society 2022** | New York | oral & poster presentations.
9. **ACS Spring 2022** | Virtual | oral presentation.
10. **APS March 2022** | Chicago | oral presentation.
11. **AIChE 2021** | Boston | oral presentation.

Awards

- University of Michigan postdoctoral travel award | ACS Fall 2024
- ACS CATL ChemCatBio travel award | ACS Spring 2023
- Poster presentation 1st place | Pittsburg-Cleveland Catalysis Society 2022
- FGSA travel award | APS March 2022
- Departmental best qualifying exam | Pennsylvania State University 2019

Broader Impact Activities

- Volunteered at the Midwest Thermodynamics and Statistical Mechanics Conference in Ann Arbor, 2024.
- Guest-spoke at the University of Michigan Center for Simulation and Data Intensive Research in Chemical Engineering, 2024.
- Poster-judged for University of Michigan Graduate Research Symposium 2023.
- Founded and led a Vietnamese Graduate Student Association at Pennsylvania State University.
- Assisted newly admitted ChE graduate students in Pennsylvania State University departmental mentorship program.